

LAB # 11



SUBMITTED TO: MA'AM FARYAL

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SECTION: CA

TASK #01

The screenshot shows the Visual Studio IDE with a C++ project named LAB. The code in LAB.cpp is as follows:

```
23 {
24     cout << "Enter Name: ";
25     cin >> name;
26
27     cout << "Enter Age: ";
28     cin >> age;
29
30     cout << "Enter Roll No: ";
31     cin >> rollNo;
32
33     cout << "Enter Department: ";
34     cin >> department;
35 }
36
37 // display() function
38 void display()
39 {
40     cout << "\nStudent Information" << endl;
41     cout << "Name: " << name << endl;
42     cout << "Age: " << age << endl;
43     cout << "Roll No: " << rollNo << endl;
44     cout << "Department: " << department << endl;
45 }
46
47 int main()
48 {
49     // ... (code for input) ...
50     display();
51 }
```

The console output shows the program running and displaying student information for ISMAIL:

```
Enter Name: ISMAIL
Enter Age: 22
Enter Roll No: 21574
Enter Department: SE

Student Information
Name: ISMAIL
Age: 22
Roll No: 21574
Department: SE

-----
Process exited after 16.45 seconds with return value 0
Press any key to continue . . .
```

The compilation results show 0 errors and 0 warnings, with an output file of C:\Users\Ismail Rahim\Desktop\LAB#11\LAB.exe.

TASK #02

The screenshot shows the Visual Studio IDE with a C++ project named LAB. The code in LAB.cpp is as follows:

```
33 rollNo = r;
34 department = d;
35 }
36
37 // Display function
38 void display()
39 {
40     cout << "Student Information" << endl;
41     cout << "Name: " << name << endl;
42     cout << "Age: " << age << endl;
43     cout << "Roll No: " << rollNo << endl;
44     cout << "Department: " << department << endl;
45 }
46
47 int main()
48 {
49     // Object creation
50     Student s1("Ali", 20, 101, "Computer Science");
51
52     // Display data
53     s1.display();
54
55     return 0;
56 }
```

The console output shows the program running and displaying student information for Ali:

```
Student Information
Name: Ali
Age: 20
Roll No: 101
Department: Computer Science

-----
Process exited after 0.2501 seconds with return value 0
Press any key to continue . . .
```

The compilation results show 0 errors and 0 warnings, with an output file of C:\Users\Ismail Rahim\Desktop\LAB#11\LAB.exe.

TASK#03

The screenshot shows an IDE with a C++ file named LAB.cpp. The code defines a `Teacher` struct with attributes `name`, `age`, `subject`, and `salary`. A `display` function is defined to print these attributes. In the `main` function, a `Teacher` object `t1` is created with the values "Sir Ahmed", 40, "Mathematics", and 75000. The `display` function is called on `t1`, and the program returns 0.

```
33     subject = s;\n34     salary = sal;\n35 }\n36 \n37 // Display function\n38 void display()\n39 {\n40     cout << "Teacher Information" << endl;\n41     cout << "Name: " << name << endl;\n42     cout << "Age: " << age << endl;\n43     cout << "Subject: " << subject << endl;\n44     cout << "Salary: " << salary << endl;\n45 }\n46 }\n47 \n48 int main()\n49 {\n50     // Create Teacher object\n51     Teacher t1("Sir Ahmed", 40, "Mathematics", 75000);\n52 \n53     // Display teacher information\n54     t1.display();\n55 \n56     return 0;\n57 }
```

The output window shows the following text:

```
Teacher Information\nName: Sir Ahmed\nAge: 40\nSubject: Mathematics\nSalary: 75000\n\n-----\nProcess exited after 0.2624 seconds with return value 0\nPress any key to continue . . .
```

TASK#04

The screenshot shows an IDE with a C++ file named LAB.cpp. The code defines a `Student` struct with attributes `name`, `age`, `rollNo`, and `department`. A `display` function is defined to print these attributes. In the `main` function, two `Student` objects, `s1` and `s2`, are created with the values "Ali", 20, 101, "Computer Science" and "Ahmed", 22, 102, "Software Engineering" respectively. The `display` function is called on both objects, and the program returns 0.

```
36 \n37 // Display function\n38 void display()\n39 {\n40     cout << "Student Information" << endl;\n41     cout << "Name: " << name << endl;\n42     cout << "Age: " << age << endl;\n43     cout << "Roll No: " << rollNo << endl;\n44     cout << "Department: " << department << endl;\n45     cout << "-----" << endl;\n46 }\n47 }\n48 \n49 int main()\n50 {\n51     // Create two Student objects\n52     Student s1("Ali", 20, 101, "Computer Science");\n53     Student s2("Ahmed", 22, 102, "Software Engineering");\n54 \n55     // Display data for both students\n56     s1.display();\n57     s2.display();\n58 \n59     return 0;\n60 }
```

The output window shows the following text:

```
Student Information\nName: Ali\nAge: 20\nRoll No: 101\nDepartment: Computer Science\n-----\nStudent Information\nName: Ahmed\nAge: 22\nRoll No: 102\nDepartment: Software Engineering\n-----\n\n-----\nProcess exited after 0.2668 seconds with return value 0\nPress any key to continue . . .
```

